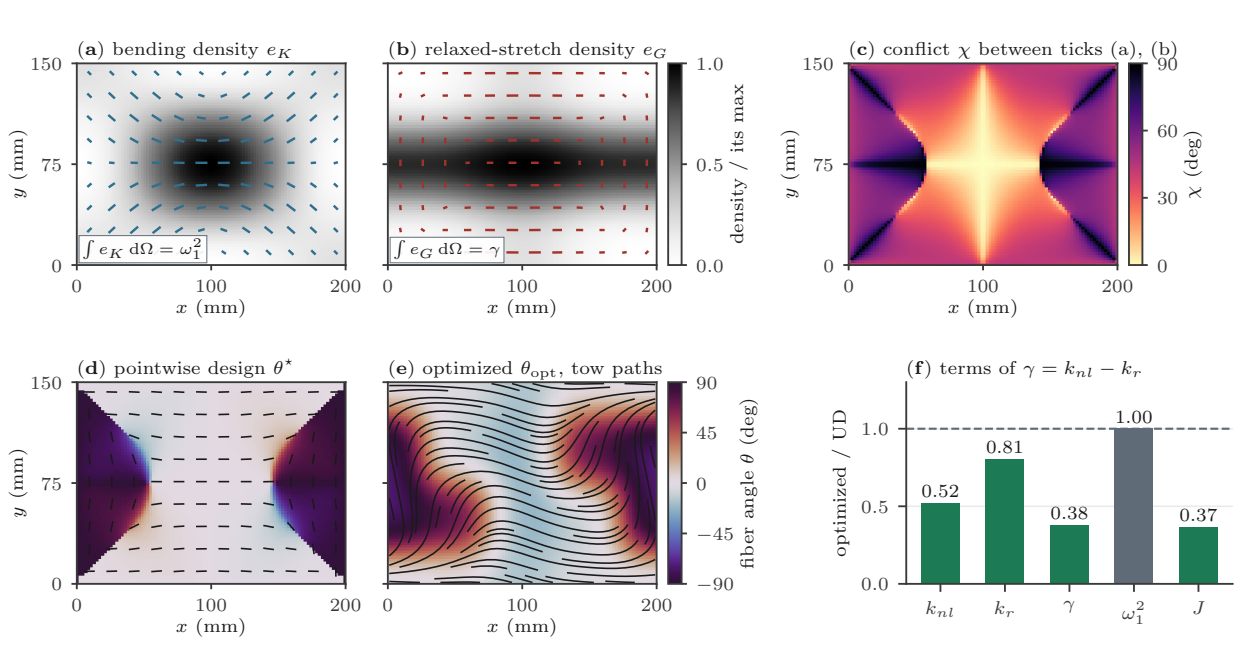

(a) bending density e_K
(b) relaxed-stretch density e_G
(c) conflict χ between ticks (a), (b)
(d) pointwise design $\theta^\star$
(e) optimized θ_{opt}, tow paths
(f) terms of $\gamma = k_{nl} - k_r$
$\int e_K \, \mathrm{d}\Omega = \omega_1^2$
$\int e_G \, \mathrm{d}\Omega = \gamma$
density / its max
χ (deg)
fiber angle θ (deg)
optimized / UD
x (mm)
y (mm)
0.52
0.81
0.38
1.00
0.37
k_{nl}
k_r
γ
ω_1^2
J